RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic Year: 2025-2026 Even

Department of Artificial Intelligence and Machine Learning

Date: 30-05-2026	Test – II	Max. Marks: 50
Semester: VI	UG	Duration: 120 minutes
Course Title: BIG DATA TECHNOLOGIES		Course Code: AI362IA

PART – A

Q. No	Question	M	BT	CO
1.1	Define book keeping objects.	2	1	1
1.2	Identify the default memory assigned to the map and reducer task?	2	1	1
1.3	State the two index types used in Hive QL?	2	1	1
1.4	List two difference between Traditional Databases and Hive .	2	2	1
1.5	Differentiate between managed table and external table in Hive.	2	2	1

PART – B

Q. No	Question	M	BT	CO
1a	Discuss the Anatomy of a MapReduce Job Run with neat figure.	7	2	1
1b	How does Hadoop estimate progress for reduce tasks when reducers process data in multiple phases?	3	2	1
2	Discuss any two failures in MapReduce jobs.	10	2	1
3a	How does bucketing work in Hive ? Illustrate with a sample use case .	4	3	1
3b	Draw and explain the architecture of Hive . Include the roles of driver , Compiler , Execution Engine and Metastore .	6	2	1
4a	Explain sorting and aggregation concepts in Hive with an example .	5	3	2
4b	Discuss how Hive organizes tables into partitions.	5	2	2
5	A **bookstore management system** maintains information about its customers, including customer ID, customer name, and city. The store also keeps records of all books available, identified by book ID, book title, and author name. Orders placed by the store are tracked using an order ID, along with publisher ID, book ID, and the quantity ordered. Details of all publishers who supply the books are maintained, including publisher ID and publisher name. Finally, the borrowing information is recorded with the customer ID, book ID, borrow date, and number of days borrowed. For the above scenario, perform the following: a. Create relevant tables b. Load the data into the tables c. Write the following queries using HiveQL i. List the books that have more than 10 copies ordered. ii. Which customers borrowed books for 7 days or more? iii. Which books are not currently available in the inventory? iv. Which book was borrowed for the longest duration (by any customer)?	10	3	3

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Academic Year: 2025-2026 Even

Department of Artificial Intelligence and Machine Learning

Date: 17-06-2026	Test – III	Max. Marks: 50
Semester: VI	UG	Duration: 120 minutes
Course Title: BIG DATA TECHNOLOGIES		Course Code: AI362IA

PART – A

Q. No	Question	M	BT	CO
1.1	List any two delivery guarantees provided by Flume.	2	1	1
1.2	What are wide transformations in Job spark? How do they influence in job status	2	2	1
1.3	State two spark's processing model.	2	1	1
1.4	Define Interceptor.	2	1	1
1.5	State two categories of operations on RDDs in Spark.	2	1	1

PART – B

Q. No	Question	M	BT	CO
1a	Explain any two methods for creating Resilient Distributed Datasets (RDDs) in Apache Spark.	5	2	1
1b	How is data serialization handled in Apache Spark?	5	2	1
2a	Discuss, with a diagram and code snippet, the working of Flume.	6	3	2
2b	Justify the Statement. "The point of Flume is to deliver large amounts of data into a Hadoop data store".	4	4	3
3a	Explain a two-tier Flume configuration using a spooling directory source and an HDFS sink.	6	2	1
3b	Discuss Integrating Flume with applications.	4	3	2
~~4a~~	~~Explain a two-tier Flume configuration using a spooling directory source and an HDFS sink.~~	6	2	1
4b	Summarize reduceByKey() and foldByKey() transformations for aggregating RDDs with an example for each	4	2	2
5	Describe the Job Submission phase of a Spark job execution. Illustrate the process using a neat diagram and explain the sequence of operations involved.	10	2	1

4a) Explain Shared variable ~~path~~. Discuss working of broadcast variable & accumulators with suitable ex. 6M.